

Leapfrogging of multiple coaxial viscous vortex rings

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A recent theoretical study [Borisov, Kilin, and Mamaev, “The dynamics of vortex rings: Leapfrogging, choreographies and the stability problem,” *Regular Chaotic Dyn.* **18**, 33 (2013); Borisov *et al.*, “The dynamics of vortex rings: Leapfrogging in an ideal and viscous fluid,” *Fluid Dyn. Res.* **46**, 031415 (2014)] shows that when three coaxial vortex rings travel in the same direction in an incompressible ideal fluid, each of the vortex rings alternately slips through (or leapfrogs) the other two ahead. Here, we use a lattice Boltzmann method to simulate viscous vortex rings with an identical initial circulation, radius, and separation distance with the aim of studying how viscous effect influences the outcomes of the leapfrogging process. For the case of two identical vortex rings, our computation shows that leapfrogging can be achieved only under certain favorable conditions, which depend on Reynolds number, vortex core size, and initial separation distance between the two rings. For the case of three coaxial vortex rings, the result differs from the inviscid model and shows that the second vortex ring always slips through the leading ring first, followed by the third ring slipping through the other two ahead. A simple physical model is proposed to explain the observed behavior. © 2015 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4915890>]

Vortex rings have been a subject of interest in vortex dynamics because of their frequent appearance in nature and technology.^{1–4} Moreover, they can be easily produced experimentally.⁵ The study of the interaction of multiple vortex rings has also attracted considerable attention^{4–9} because of its relevance to the fundamental understanding of vortex-vortex interaction that occurs in many unsteady flow phenomena such as jets and plumes.

Past studies^{1–15} have shown that when two identical coaxial vortex rings travel in the same direction, it could lead to a process often referred to as leapfrogging of vortex rings. During this process, the induced velocity of the leading ring causes the rear ring to contract radially and accelerate, and concurrently, the induced velocity of the rear ring causes the leading ring to expand and slow down. The rear ring eventually catches up with the leading ring and slips through it, and the process repeats until they merge into a single ring. Although the leapfrogging of two vortex rings has been extensively investigated experimentally, numerically, and theoretically for many years (see an excellent review by Meleshko⁶), the case involving more than two vortex rings has received little attention. In a recent theoretical study based on topological approach and concept of a bifurcation, complex leapfrogging of three inviscid vortex rings was reported for the first time.¹⁶ The result reveals that each of the vortex rings alternately slips through the other two rings ahead. They also showed the similar leapfrogging of three viscous vortex rings with the same initial circulation, different radii, and different separation distances by solving the axisymmetric Navier-Stokes equations.¹⁷ These interesting observations raise an obvious question: do viscous vortex rings with an identical initial circulation, radius, and separation distance also undergo the same leapfrogging process? The desire to answer this question motivates the

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present study. This study also provides an opportunity to determine the flow conditions under which successful leapfrogging of two vortex rings can be achieved or not. As far as we are aware, this has not been carried out previously. To study the motion of multiple coaxial viscous vortex rings, the lattice Boltzmann method (LBM), which has been previously verified and applied to vortex dynamics,^{18–20} is used to simulate the motion of multiple viscous rings.

We consider multiple coaxial viscous rings moving in the same direction. The rings have identical initial circulation Γ_0 , radius R_0 , and core size σ_0 . The separation distance between neighboring rings is set initially at h_0 (see Fig. 1(a)) and the vorticity distribution of each ring is assigned by a Gaussian function $\omega_\theta = \Gamma_0 e^{-(r/\sigma_0)^2}/(\pi\sigma_0^2)$, where r is the radial distance from the center of the core and ω_θ is the azimuthal vorticity. The computational domain is $l_x \times l_y \times l_z$, where l_x , l_y , and l_z are the dimensions of the domain in the x , y , and z directions, respectively. The domain is discretized by a uniform structure mesh. The periodic boundary condition is used at the boundaries.

The numerical method employed is the same as the one used in Ref. 18. The LBM method can be seen as a finite difference method with central-differencing and explicit time integration^{21,22} and it is formally second order accurate in space.²³ The code has been extensively validated for several benchmark problems involving the motion of a vortex ring in viscous liquid.^{18–20} Interested readers are referred to Refs. 18–20 for the implementation details. Here, we check the grid sensitivity on the present computation. For this purpose, we simulate the motion of a single ring ($\sigma_0 = 0.1R_0$) in a viscous fluid. A grid convergence study is carried out using different grid densities defined by the initial ring radius, $R_0 = 25\Delta x, 50\Delta x, 100\Delta x, 200\Delta x$, and $250\Delta x$ for $Re = 500$ and 5000 , where $Re = \Gamma_0/\nu$ and ν is the kinematic viscosity, Δx is grid size. Fig. 1(b) shows the influence of grid resolution on the enstrophy of flow field. The enstrophy is defined by $E_\Omega = 0.5 \iint \Omega \cdot \Omega dv$, where Ω is the curl of the velocity ($\mathbf{u}$). Although the values of E_Ω increase with increasing grid density, the time history curves of E_Ω obtained with the grid density $R_0 = 50\Delta x$ and $R_0 = 100\Delta x$ at $Re = 500$ and $R_0 = 200\Delta x$ and $R_0 = 250\Delta x$ at $Re = 5000$ almost coincide. The present results show that the resolution $R_0 = 200\Delta x$ is sufficient for the vortex ring simulation of $Re \leq 5000$. To provide further verification, using the grid density $R_0 = 200\Delta x$, we simulate leapfrogging of two coaxial rings which has been studied both numerically and experimentally.^{5,9} In Fig. 2, we compare results obtained by the present method with those computed by the spectral method⁹ at $Re = 1000$. It is clear that both LBM and spectral methods produce almost the same flow structures of ring-ring interaction at the same Reynolds number. To investigate the influence of the azimuthal instability of vortex rings, we added a sine wave perturbation as used in other numerical simulations²⁴ in the z -direction, with an amplitude $\epsilon = 0.05R_0$ and the wavenumber $n = 8$. In Fig. 3, we compare our numerical results with and without the perturbation with the experimental observation⁵ for $Re = 3415$. It can be seen that the salient flow features associated with the ring-ring interaction are reproduced in our simulations. Although the perturbation can hasten the merging of the two rings (see Fig. 3(b)), it does not change the essential leapfrogging features. In this paper, all things considered, a sufficiently large computational domain $l_x \times l_y \times l_z = 7R_0 \times 7R_0 \times 20R_0$ ²⁴ and the higher resolution $R_0 = 200\Delta x$ ²⁰ are adopted for the subsequent computations. To simplify the problem, we exclude the influence of flow instability in the

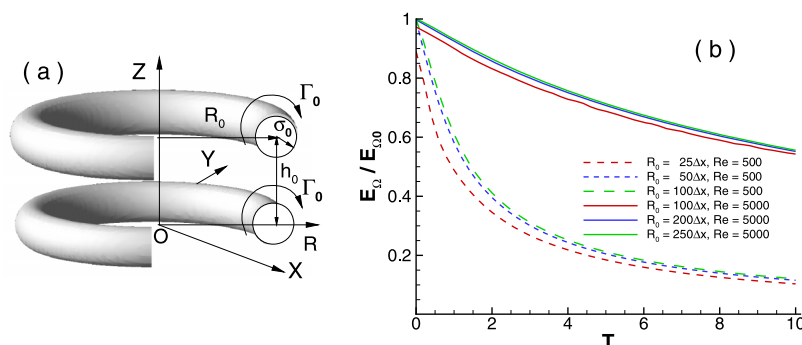


FIG. 1. (a) Schematic of vortex rings configuration used in the present study. (b) The influence of mesh size defined by the initial ring radius on the enstrophy of flow field induced by a single vortex ring with $\sigma_0 = 0.1R_0$ at $Re = 500$ and 5000 .

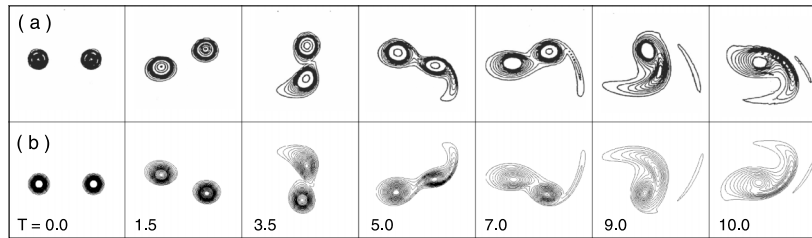


FIG. 2. Leapfrogging of two vortex rings with $\sigma_0 = 0.1R_0$ and $Re = 1000$. Comparison of the vorticity contours in the $Z - R$ plane of symmetry from (a) the spectral method⁹ and (b) the present numerical method ($R_0 = 200\Delta x$).

calculations. The computations are conducted for $0.4R_0 \leq h_0 \leq 4R_0$, $200 \leq Re \leq 5000$, $\sigma_0 = 0.1R_0$, and $0.2R_0$. R_0 and Γ_0/R_0^2 are used as the reference length and time, respectively.

We first investigate the well-studied case of leapfrogging between two identical coaxial vortex rings. The computed results depicted in Figs. 4 and 5 show the time-evolution of the core shape and its center vorticity for different Reynolds numbers. They reveal two obvious stages of vortex interaction, namely, leapfrogging stage and merging stage. In the leapfrogging stage, the vortex core of the leading ring maintains a rounder shape, whereas the core of the rear ring is more elliptic in shape. Once the rear ring slips through the leading ring, their roles are reversed. In the merging stage, the process is characterized by larger core deformations of the slip-through ring, which is subsequently entrained by the leading ring. It is of interest to note from Fig. 4 that the number of successful leapfrogging cycles is very much dependent on Reynolds number of the rings. Here, only one cycle of leapfrogging is observed for $Re = 1000$ and two cycles for $Re = 3000$ before the two rings merged into one. This finding differs from the inviscid case ($Re \rightarrow \infty$), where the leapfrogging process repeats continually. To examine the state of the core during the interaction, computed time evolution of the vorticity at the core center is compared with that of a single ring as shown in Fig. 5(a). The result shows that for a given Reynolds number, the core center vorticity of both the leading and rear rings decreases with time, with the rate of decline higher for the rear ring than the leading ring. Closer examination of the result in Fig. 5(a) shows that prior to leapfrogging, the leading ring has a higher core centre vorticity than an isolated vortex ring, whereas the rear ring has a lower core centre vorticity. This finding is consistent with the fact that radial expansion reduces core area (in order to satisfy continuity) and

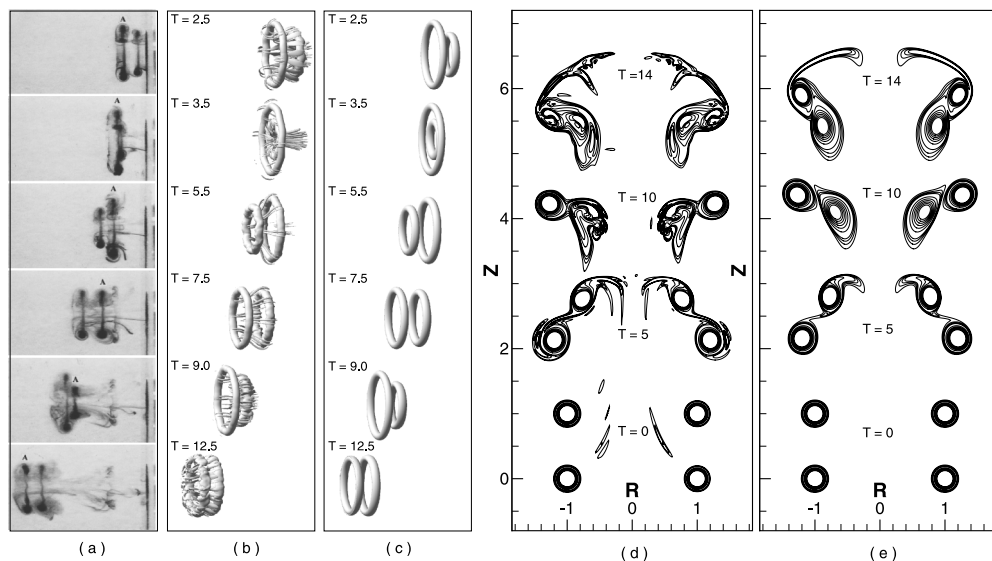


FIG. 3. Comparison of the development of flow structures from (a) experimental flow visualization⁵ and (b) the present numerical method with the perturbation. (c) The present numerical method without the perturbation for $Re = 3415$ and $\sigma_0 = 0.1R_0$ at different times. Comparison of vorticity contours in the $R - Z$ plane obtained from the present method (d) with the perturbation and (e) without the perturbation.

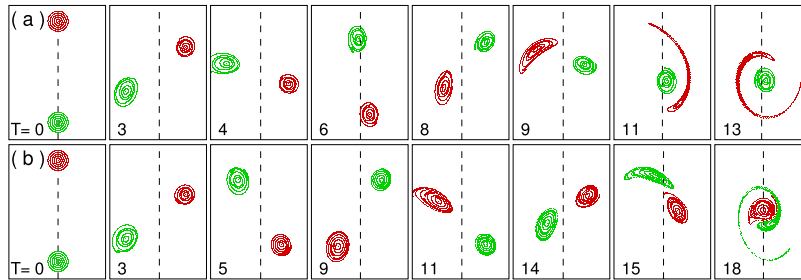


FIG. 4. Core shapes of two coaxial vortex rings at different stages of interaction with $h_0 = R_0$, $\sigma_0 = 0.1R_0$, and different Re . The core is described by trajectories of fluid particles. (a) $Re = 1000$ and (b) $Re = 3000$.

intensifies core vorticity, whereas radial contraction increases core size and weakens core vorticity. The higher rate of decline of vorticity for the rear ring could be due to the radial contraction bringing vortex cores of opposite circulation closer together, thus promoting vorticity annihilation in the region around the axis of symmetry. This would explain why the contracting rear ring is always engulfed by a relatively stronger expanding leading ring during the merging process (see Fig. 4). The results in Fig. 5(b), which depicts core center trajectories of vortex rings in the $Z - R$ plane and typical vortex structures at selected instants, show that irrespective of the Reynolds number, two coaxial viscous vortex rings eventually merge to form a single ring. Also, larger initial separation distance (h_0) leads to an expected lower frequency of leapfrogging. To demarcate the conditions under which leapfrogging can be achieved, the results are presented in $Re - h_0$ parameter space for two different core sizes as shown in Fig. 6. Also included in the figure are selected typical vortex structures of the first passage. The results show that successful leapfrogging is a function of not only the Reynolds number but also of the initial vortex core size (σ_0) and initial separation distance (h_0) between the two rings. The results further show that for a given h_0 , larger σ_0 requires higher critical Reynolds number to achieve successful leapfrogging. This observation can be attributed to substantial distortion of a larger core (which has more substantial distortion than a smaller core), coupled with vorticity annihilation during the slip-through process as discussed earlier. In addition, for a given core size, an increase in Reynolds number can also produce successful leapfrogging at higher h_0 . This is not unexpected since higher Re translates into higher kinetic energy which encourages leapfrogging. The result depicted in Fig. 6 may help to explain why some past experiments could achieve successful leapfrogging,⁵ while others could not (Maxworthy 1972⁷ and Oshima *et al.* 1975²⁵). Unfortunately, the results of the latter studies could not be incorporated in Fig. 6 because the values of vortex core sizes were not provided by the authors.^{7,25} It should be mentioned that in real case, azimuthal instability of the rings will change the flow dynamics dramatically.

We now study the motion of three and four coaxial vortex rings. The results of the core shapes and core trajectories are shown in Fig. 7 for $h_0 = R_0$ and $Re = 3000$. In the case of the three identical vortex rings with Γ_0 , R_0 , and h_0 traveling in the same direction, their mutual induction causes the leading or

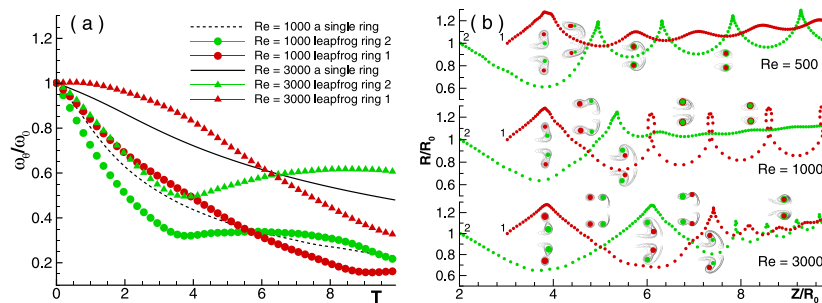


FIG. 5. (a) The time-evolution of core center vorticity of rings (the intersection point of the vorticity curves for the leading and rear ring corresponds to the instant the rear ring is in the midst of slipping through the leading ring and the four vortex cores are almost aligned vertically), and (b) trajectories of the core centers in the $Z - R$ plane and vortex structures at different stages of the interaction with $h_0 = R_0$, $\sigma_0 = 0.1R_0$, and different Re .

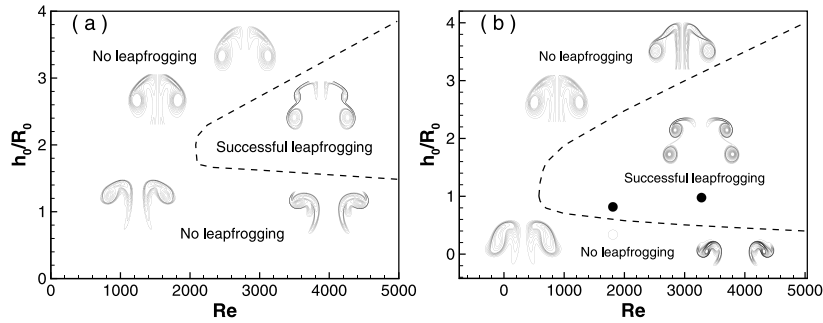


FIG. 6. Vortex structures in the $Re-h_0$ space. The characteristic structure is represented by vorticity contours at the stage of the rear ring passing through or merging with the front ring. (a) $\sigma_0 = 0.2R_0$ and (b) $\sigma_0 = 0.1R_0$. The symbols are taken from experiments. •—successful leapfrogging and o—no leapfrogging.

first ring to expand radially and slow down, while the second and third vortex rings contract and accelerate. The radial contraction of the second ring is less than that of the third ring because the induced velocity of the first ring on the second ring is offset by the induced velocity of the third ring on the second ring. On the other hand, the third ring experiences the combined effect of the induced velocity of the first and second rings, resulting in greater radial contraction and acceleration. Moreover, the slowdown of the first ring (due to the combined effect of the second and third rings) helps to promote the second ring to slip through it. As the third ring continues to contract and accelerate, it eventually catches up with the two leading rings and passes through the first ring and then the second ring. The order of the passage of the vortex rings clearly differs from the prediction of inviscid model¹⁶ in which each of the rings alternately slips through the other two. Obviously, viscous effect has contributed to the difference in the results. As discussed earlier, a contracting ring brings the vortex cores of opposite

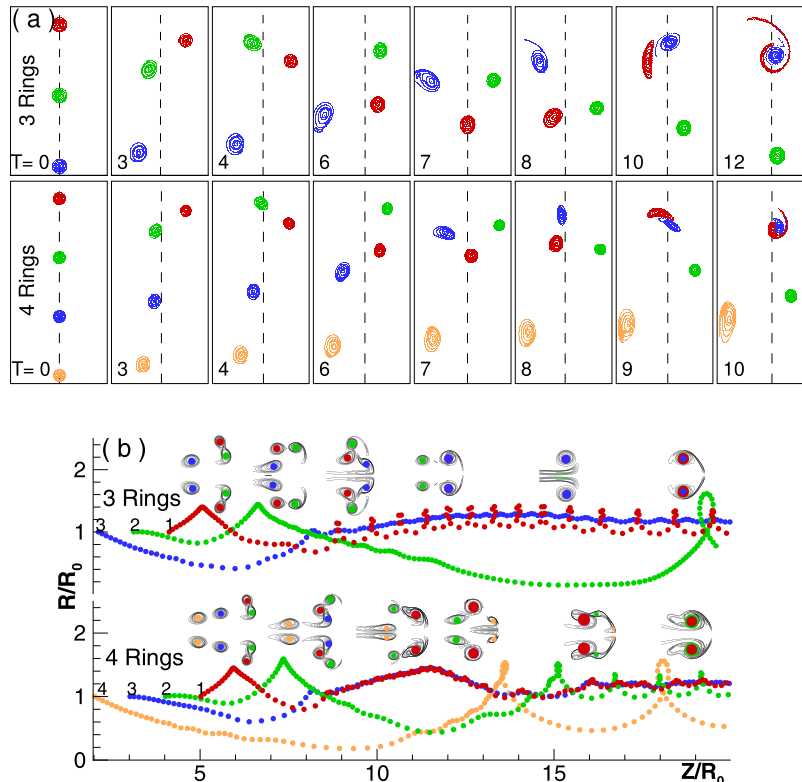


FIG. 7. (a) Core shapes of multiple coaxial rings interaction and (b) trajectories of the core centers of rings in the $Z-R$ plane and vortex structure at different stages of the interaction with $h_0 = R_0$, $\sigma_0 = 0.1R_0$, and $Re = 3000$.

circulation closer together, thus promoting vorticity annihilation and lowering its circulation. This translates into lower self-induced velocity and induced velocity of the contracting ring on the neighboring rings. A similar behavior was observed in the four-ring interaction (see Fig. 7). In all cases, the final outcome of the interaction, whether it is two, three, or four co-axial vortex rings, eventually was the merging into a single vortex ring.

Based on the above results, we now propose a simple physical model to explain the observed leapfrogging behavior of multi-coaxial vortex rings and why they differ from the inviscid model. Using three identical coaxial vortex rings of radius R_0 and axial separation $h_0 = \Delta Z_0$ as an example, each ring moves forward with the sum of its self-induced velocity U_S and induced velocity U_I due to the other rings, which can be estimated³ by $U_I = \Gamma_0[1 - e^{-(r/\sigma)^2}]/(2\pi r)$ and $U_S = \Gamma_0[\ln(8R/\sigma) - 0.558]/(4\pi R)$. In our model, we assume (1) the volume of ring core is constant and (2) the circulation around the ring changes with its core size and time, i.e., $\Gamma = \Gamma_0[1 - \chi e^{-\sigma^2/(\gamma t)}]$, $\chi = [\sigma(t) - \sigma(t - \Delta t)]/\sigma(t - \Delta t)$, ($\gamma \geq 0, R > \sigma$), where γ acts as a viscous influence coefficient. The translation velocity of every ring at time t can be written as $\mathbf{U}_i = U_i \mathbf{e}_r + V_i \mathbf{e}_z = U_{Si} \mathbf{e}_z + \sum_{j \neq i}^{j=3} U_{Iji} \sin \theta_{ji} \mathbf{e}_r - U_{Iji} \cos \theta_{ji} \mathbf{e}_z$, where $U_{Iij} = \Gamma_i[1 - e^{-(r_{ij}/\sigma_i)^2}]/(2\pi r_{ij})$, $U_{Si} = \Gamma_i[\ln(8R_i/\sigma_i) - 0.558]/(4\pi R_i)$, $\sin \theta_{ij} = (Z_j - Z_i)/r_{ij}$, and $\cos \theta_{ij} = (R_j - R_i)/r_{ij}$. The displacement of vortex core center can be calculated by $R_i = R_{i0} + \int_0^t U_i dt$ and $Z_i = Z_{i0} + \int_0^t V_i dt$.

Figure 8 depicts time-variation of vortex ring radii and their displacement in the z -direction for the given initial conditions ($h_0 = R_0 = 2$ m, $\Gamma_0 = 2$ m²/s, and $\sigma_0 = 0.1R_0$) at different γ , bearing in mind that γ increases with viscosity. When $\gamma = 0$ (i.e., no viscosity), the velocity of tail ring translates much faster than the second ring (see Fig. 8(a)). It eventually catches with the second ring, slips through it, and then through the leading ring. Therefore, in the absence of viscosity, our model agrees with the inviscid prediction.¹⁶ When $\gamma > 0$, the circulation of the ring is affected by the combined effect of core distortion, viscous diffusion, and vorticity annihilation. When $\gamma < 0.0053$, the viscous effect is too small to influence the order of slip-through of the vortex rings. However, when $\gamma \geq 0.0053$, the viscous effect becomes more significant. Since the tail ring has a much larger core deformation than the second ring, its circulation weakens considerably as it undergoes radial contraction and vorticity annihilation as discussed earlier. This translates into a reduced self-induced velocity U_S and induced velocity U_I of tail ring. This feature is more obvious at higher γ , as shown in Figs. 8(b) and 8(c). The

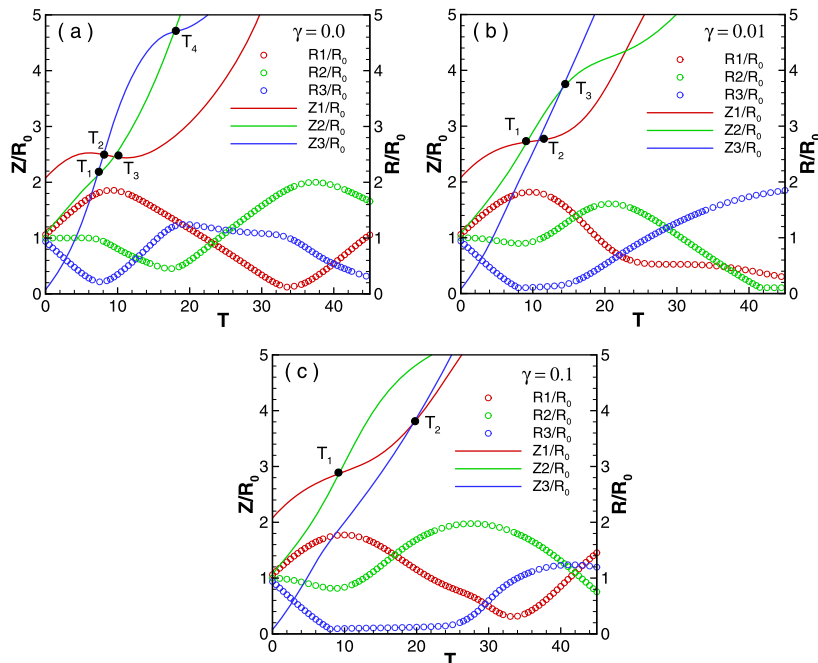


FIG. 8. The variation of the displacement (Z_i) and radius (R_i) of three vortex rings with time (T) for different γ .

outcome of this study shows that when the viscous effect is not negligible, the second ring will first slip through the leading one, which is consistent with our numerical simulation and further supports our finding.

In conclusion, we have numerically simulated the interaction of multiple identical coaxial vortex rings for a range of flow conditions, and the results show that viscous effect dictates not only the number of successful leapfrogging but also the order of passage of vortex rings. For the case of two coaxial vortex rings, our results plotted in $Re - h_0$ parameter space provide useful information on the conditions for successful leapfrogging, and this may help to explain why some past experiments could achieve successful leapfrogging of two vortex rings while the others could not. For more than two viscous vortex rings, the present results show that the second ring slips through the leading one, followed by the third ring slipping through the other two ahead. This is contrary to the inviscid flow prediction, which shows that each of the rings alternately slips through the other two. To elucidate the physics of the phenomenon, we proposed a simple physical model, based on vorticity dissipation of the deformed vortex core of the contracting ring, to explain the observed flow behavior revealed in the simulation.

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